Angular Momentum as Information Geometry: Isotropic Packing and the Universal Angular Cross-Section

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[SYMBOLIC PROOF] We show that the angular momentum quantum numbers (ℓ, m) emerge from a purely geometric construction: uniform-density packing of distinguishable states on concentric isotropic shells in two dimensions. Starting from two axioms—discrete binary distinguishability and geometric indifference (maximum entropy)—the construction yields per-shell angular capacities $2k-1$ for shell k, which is isomorphic to the spherical harmonic basis $Y_{\ell m}$ with $\ell = k-1$ and $m = -\ell, \ldots, +\ell$. The principal quantum **[OBSERVATION]** number n corresponds to a cumulative grouping of the first n shells, producing the n^2 orbital count, but this grouping is not forced by the axioms alone. Similarly, a factor of 2 arises from the orientability of the compactified packing manifold (S^2), corresponding to spin multiplicity. When the packing is interpreted as a graph, its Laplacian splits into ℓ-blocks (the edges change n or m, never ℓ), and **[MEASURED]** its top eigenvalue saturates the bipartite bound $\lambda_{\max} \to 2d_{\max} = 8$ (attained in a mid-ℓ block, not in the s-wave block, whose own bound is 4; corrected 2026-09-03), so that under a single matched scale $K_{\text{vac}} = -1/16$ the extremal eigenvalue reaches the hydrogen ground-state energy—matched, not derived: the content that could have failed is the spectral bound, not the target the scale is matched to; the full per-shell spectrum $E_n = -Z^2/(2n^2)$ requires an additional node-weight term (Paper 18 [13] § III). **[OBSERVATION]** This angular cross-section recurs as a universal structural factor inside every natural geometry developed in the Geometric Vacuum series—from S^3 for atoms through composed fiber bundles for multi-electron molecules—making it a foundational motif of the framework. This is Paper 0 in the series.

INTRODUCTION

The hydrogen atom possesses n^2 orbital states per principal quantum number n, each labeled by angular momentum $\ell = 0, 1, \ldots, n-1$ and magnetic quantum number $m = -\ell, \ldots, +\ell$, with a spin multiplicity of 2 giving $2n^2$ states total. The standard derivation obtains these labels from the eigenvalue structure of the angular momentum operators $\hat{L}^2$ and $\hat{L}_z$ acting on the Hilbert space of the Schrödinger equation. Here we ask: to what extent can this quantum number structure be recovered from geometry alone, without invoking operators, Hilbert spaces, or the eigenvalue equation?

We present a construction based on uniform-density packing of distinguishable states on concentric isotropic shells in two dimensions. The construction rests on two axioms: (i) a minimum of two distinguishable states is required to define a metric (binary distinguishability), and (ii) in the absence of external constraints, the distribution of states must maximize entropy, enforcing isotropy (geometric indifference). These axioms, together with the natural choices of concentric shells and area-proportional state counting, produce the following results at three distinct levels of strength.

[SYMBOLIC PROOF] Strong claim: angular quantum numbers. Shell k has capacity $2k-1$, which is isomorphic to the spherical harmonic basis $Y_{\ell m}$ with $\ell = k-1$. The angular momentum labels (ℓ, m) and the per-shell degeneracy $2\ell+1$ emerge directly from the packing. This is the central, mathematically rigorous result of the paper.

[OBSERVATION] Structural correspondences. The principal quantum number n arises when one groups the first n shells cumulatively, yielding the count $\sum_{k=1}^{n}(2k-1) = n^2$. This grouping is natural but not uniquely forced by the axioms. The factor of 2 (spin multiplicity) arises from the orientability of the compactified packing manifold: conformally compactifying the plane to S^2 introduces two orientations, corresponding to a $\mathbb{Z}_2$ **[OBSERVATION]** doubling. The identification of this doubling with spin-$\frac{1}{2}$ is a structural correspondence, not a derivation from the axioms.

[OBSERVATION] Series significance. The (ℓ, m) angular sector constructed here recurs as a universal structural factor inside every natural geometry developed in the Geometric Vacuum (GeoVac) series: S^3 for hydrogen [6], prolate spheroidal coordinates for H_2^+ [7], hyperspherical coordinates for He [8], molecule-frame hyperspherical coordinates for H_2 [9], and composed fiber bundles for LiH [10]. In each case, the full geometry separates into a system-specific radial or hyperradial part and a universal angular cross-section whose structure is precisely the packing derived here. This universality is the reason the packing construction is foundational to the series rather than merely a curiosity about hydrogen.

Several elements of this work connect to well-established results. The counting $\sum_{\ell=0}^{n-1}(2\ell+1) = n^2$ is standard SO(3) representation theory. The relationship between hydrogen's "accidental" degeneracy and the geometry of S^3 was established by Fock [1] and Bargmann [2]. What is new here is the specific packing algorithm that produces the (ℓ, m) labeling from geometric–combinatorial reasoning independent of the operator formalism, and the observation that this angular sector is the universal fiber across the GeoVac natural geometries.

The paper is organized as follows. Section presents the packing algorithm. Section establishes the isomorphism between shell capacity and the spherical harmonic basis. Section discusses the structural correspondences for n and spin. Section shows how the angular cross-section recurs across the GeoVac natural geometries. Section constructs the graph topology and connects it to the spectral results of later papers. Sections and address scope, limitations, and context.

THE PACKING CONSTRUCTION

Axioms

The construction rests on two axioms, which contain no reference to quantum mechanics, operators, wavefunctions, or physical observables.

Axiom 1 (Binary Distinguishability). To define a metric space, a minimum of two distinguishable states ($N_{\text{init}} = 2$) is required to establish a non-zero fundamental scale d_0. A single point cannot define a distance; two points create the first bit of spatial information.

Note: The choice $N_{\text{init}} = 2$ is the minimal non-trivial initialization. It is the simplest choice consistent with this axiom, but not the only one; initializing with $N_{\text{init}} = 3$ or more points would define a different fundamental area and modify the downstream counting.

Axiom 2 (Geometric Indifference). In the absence of external constraints or privileged directions, the distribution of states must maximize entropy. By the Principle of Indifference, this necessitates isotropy: states populate isotropic shells (circles) rather than anisotropic tilings (squares, hexagons), which would imply privileged directions not present in the axioms.

Note: Isotropy motivates circular shells over anisotropic tilings, but uniform-density packing on concentric circles is one of several ways to implement isotropy—for instance, hexagonal close-packing achieves higher packing density while preserving six-fold symmetry. Concentric shells with area-proportional counting is the simplest realization consistent with the axioms, not the only one.

These axioms, together with the specific choices of concentric circular shells, area-proportional counting, and binary initialization, constitute the full set of assumptions. The axioms motivate these choices but do not uniquely determine them. What is remarkable is that this particular realization—arguably the simplest one consistent with the axioms—produces exactly the angular momentum quantum numbers.

Algorithm

Step 1. Place $N_{\text{init}} = 2$ points on an isotropic shell of radius $r_1 = d_0$, separated by the fundamental distance d_0.

Rationale: Two points are the minimum needed to define a distance (Axiom 1). Isotropy requires a circular shell (Axiom 2). This fixes both d_0 and the fundamental area per state $\sigma_0 = \pi d_0^2/2$.

Step 2. Add concentric isotropic shell k at radius $r_k = k \cdot d_0$.

Step 3. Determine the state count N_k on shell k from uniform information density. The annular area between shells $k-1$ and k is

$$A_k = \pi r_k^2 - \pi r_{k-1}^2 = \pi d_0^2\left[k^2 - (k-1)^2\right] = \pi d_0^2(2k-1)\,. \tag{1}$$

Dividing by the fundamental area gives the state count:

$$N_k = \frac{A_k}{\sigma_0} = \frac{\pi d_0^2(2k-1)}{\pi d_0^2/2} = 2(2k-1)\,. \tag{2}$$

Step 4. Repeat for $k = 1, 2, 3, \ldots$

Resulting structure

The construction yields the distribution shown in Table I.

Shell k	States $N_k = 2(2k-1)$	Cumulative $\sum_{i=1}^{k} N_i$
1	2	2
2	6	8
3	10	18
4	14	32
5	18	50
$\vdots$	$\vdots$	$\vdots$
k	$2(2k-1)$	$2k^2$

TABLE I. State distribution on isotropic shells. Each shell k carries $2(2k-1)$ states; the cumulative count through shell k is $2k^2$.

Two features of this table merit immediate comment. First, the per-shell count $2(2k-1)$ factorizes as $2 \times (2k-1)$, where the factor $2k-1$ is an odd integer determined entirely by the area formula. Second, the cumulative count $2k^2$ matches the hydrogen degeneracy $2n^2$ if one identifies $k = n$. These observations are analyzed in the next two sections at appropriately different levels of strength.

ANGULAR QUANTUM NUMBERS FROM UNIFORM DENSITY

This section contains the central result of the paper: the per-shell state count from the packing construction is isomorphic to the angular momentum labeling (ℓ, m).

The angular capacity of shell k

The factor of $2(2k-1)$ in Eq. (2) factorizes into a global factor of 2—whose origin in the initialization condition and manifold orientability is analyzed in Sec. —and an angular capacity per shell:

$$N_k^{\text{angular}} = 2k - 1 \tag{3}$$

distinct angular slots. **[SYMBOLIC PROOF]** The annular-area derivation of this $2k-1$ angular capacity (equivalently $2\ell+1$ with $\ell = k-1$) is verified by `tests/test_trunk_qa_annular.py`.

Isomorphism with spherical harmonics

The spherical harmonics $Y_{\ell m}(\theta, \phi)$ for a given ℓ have $2\ell+1$ independent components labeled by $m = -\ell, -\ell+1, \ldots, +\ell$. Under the identification

$$\ell = k - 1\,, \tag{4}$$

the angular capacity of shell k becomes

$$N_k^{\text{angular}} = 2(k-1) + 1 = 2\ell + 1\,, \tag{5}$$

which is precisely the dimension of the spin-ℓ irreducible representation of SO(3). The $2k-1$ angular positions on shell k are in one-to-one correspondence with the magnetic quantum numbers $m = -\ell, \ldots, +\ell$.

[SYMBOLIC PROOF] This is not a coincidence that must be checked case by case: it is an algebraic identity. The area formula (Eq. 1) forces the annular area of shell k to be proportional to $2k-1$, which is the dimension of the $(k-1)$-th irreducible representation of SO(3). The packing **[SYMBOLIC PROOF]** construction therefore generates the complete angular momentum basis $\{(\ell, m) : 0 \le \ell,\ -\ell \le m \le \ell\}$ as a direct consequence of uniform density on isotropic shells.

What the construction produces

To be precise about the scope of this result:

- The construction produces the *labeling* (ℓ, m) and the *degeneracy* $2\ell+1$ per shell. These are the angular quantum numbers.
- It does *not* produce the functional form of the spherical harmonics, the eigenvalue $\ell(\ell+1)$ of $\hat{L}^2$, or the commutation relations of angular momentum operators. Those require additional structure (the graph Laplacian and its continuum limit), developed in Papers 1 and 7.
- The result is *independent* of any specific physical system. The angular labeling emerges from 2D isotropic geometry, not from the hydrogen atom. Hydrogen merely happens to use this same angular sector—as do all rotationally symmetric quantum systems.

STRUCTURAL CORRESPONDENCES: n, SPIN, AND $2n^2$

The angular quantum numbers (ℓ, m) emerge directly from the packing (Sec.). The principal quantum number n and the spin factor of 2 involve additional identifications that are natural and mathematically grounded but not uniquely forced by the two axioms. We present them here as structural correspondences.

The principal quantum number as cumulative grouping

If one groups the first n shells cumulatively, the total angular state count is

$$\sum_{k=1}^{n} (2k-1) = n^2\,, \tag{6}$$

which matches the orbital degeneracy of hydrogen's n-th principal shell: $\sum_{\ell=0}^{n-1}(2\ell+1) = n^2$. Including the spin factor gives $2n^2$.

This grouping is natural—the shells are concentrically ordered, so cumulative counting is the simplest aggregation—but the axioms do not *require* that shells be grouped this way. The packing generates an infinite sequence of shells; the decision to collect the first n of them into a single principal shell is an additional structural choice that corresponds to, but is not derived from, the axioms.

The factor of 2 and orientability

The per-shell count $N_k = 2(2k-1)$ contains a factor of 2 that traces back to two sources:

1. **Initialization.** Axiom 1 places $N_{\text{init}} = 2$ states on the first shell, fixing the fundamental area $\sigma_0 = \pi d_0^2/2$. This factor of 2 propagates through all subsequent shells via the uniform-density condition.

2. **Orientability.** When the flat packing plane is conformally compactified to the Riemann sphere $S^2 \cong \mathbb{C}^*$, the resulting manifold is orientable: it admits two distinct normal directions (inward and outward). This orientability provides a natural $\mathbb{Z}_2$ doubling of the state space. Mathematically, tracking this orientation lifts the symmetry group from SO(3) to its double cover SU(2).

[OBSERVATION] The identification of this $\mathbb{Z}_2$ doubling with the spin-$\frac{1}{2}$ degree of freedom is consistent with the mathematical structure: orientability on S^2 and spinor representations of SU(2) both produce a factor of 2. However, the packing construction does not derive the spin angular momentum algebra $[\hat{S}_i, \hat{S}_j] = i\epsilon_{ijk}\hat{S}_k$ or the eigenvalue $s(s+1)$. The correspondence is between the *counting* (a factor of 2) and spin *multiplicity* ($2s+1 = 2$ for $s = 1/2$), not between the construction and the full spinor formalism.

A remark on the axioms

The construction follows from the two axioms together with several additional structural choices: concentric shells with linearly spaced radii ($r_k = kd_0$), area-proportional state counting, and the specific fundamental area $\sigma_0 = \pi d_0^2/2$ fixed by the initialization. These choices are natural and mutually consistent—they are arguably the simplest possibilities compatible with the axioms—but they are not uniquely determined by the axioms alone. The axioms constrain the construction; they do not uniquely fix every detail of it.

This distinction matters for the epistemic status of the results. The angular quantum numbers (ℓ, m) follow from any isotropic, uniform-density, concentric construction—a robust class. The specific identification of n and the factor of 2 depend on the particular choices above.

THE PACKING AS UNIVERSAL ANGULAR CROSS-SECTION

The packing construction of Sec. produces a universal (ℓ, m) angular labeling. In the subsequent papers of the GeoVac series, this same angular sector appears as a structural factor inside every natural geometry used for quantum chemistry calculations. The full geometry of each system separates into a system-specific radial (or hyperradial) coordinate and the universal angular cross-section derived here.

Table II summarizes the hierarchy.

Level	System	Full Geometry	Angular Cross-Sec
1	H (1-center, $1e$)	S^3 (Fock projection)	$Y_{\ell m}$ on $S^2 \subset S^3$
2	H_2^+ (2-center, $1e$)	Prolate spheroidal	m exact; ℓ-like ind
3	He (1-center, $2e$)	Hyperspherical	$Y_{\ell m}$ per electron i
4	H_2 (2-center, $2e$)	Mol-frame hyperspherical	$Y_{\ell m}$ angular chann
5	LiH (core + valence)	Composed fiber bundle	$Y_{\ell m}$ per electron g

TABLE II. The (ℓ, m) angular cross-section inside each natural **[OBSERVATION]** geometry of the GeoVac series. In every case, the full coordinate system separates into a system-specific radial part and the universal angular labeling constructed in this paper.

Level 1: Hydrogen on S^3. Fock's 1935 stereographic projection [1] maps the hydrogen momentum-space problem onto the three-sphere S^3. The eigenstates of the Laplace–Beltrami operator on S^3 are hyperspherical harmonics, which decompose as $Y_{n\ell m} = R_n(\chi)\, Y_{\ell m}(\theta, \phi)$. The angular factor $Y_{\ell m}$ is precisely the per-shell structure of the packing, with $\ell = k - 1$. Paper 7 [6] shows that the continuum limit of the graph Laplacian on the $2n^2$ packing is conformally equivalent to this S^3 operator (the convergence itself is only indicated numerically, not established, through $n_{\max} = 30$).

Level 2: H_2^+ on prolate spheroids. For one-electron diatomics, the natural geometry is the prolate spheroidal coordinate system (ξ, η, ϕ), in which the Schrödinger equation separates. The azimuthal quantum number m survives exactly as the quantum number of the ϕ-equation ($e^{im\phi}$), directly inheriting the packing's angular slot labeling. The quantum number ℓ is no longer exactly conserved in the two-center problem: it is replaced by a separation parameter from the η-equation that reduces to ℓ in the united-atom limit. The angular structure is recognizably descended from the (ℓ, m) packing—m is exact, and the ℓ-like index organizes the η-eigenstates into the same degeneracy pattern—even though the two-center potential breaks the full SO(3) symmetry to axial SO(2). Paper 11 [7] constructs a discrete lattice in these coordinates.

Level 3: Helium in hyperspherical coordinates. For two-electron atoms, the six-dimensional configuration space is parameterized by a hyperradius R and five angles. The angular eigenstates decompose into single-electron angular momenta (ℓ_1, m_1) and (ℓ_2, m_2), each carrying the per-shell structure of the packing. Paper 13 [8] implements this as a fiber bundle: the base space is the hyperradius, and the fiber is the angular manifold.

Level 4: H_2 in molecule-frame hyperspherical coordinates. For two-electron diatomics, the geometry combines the molecular axis (internuclear separation R) with the hyperspherical treatment of the two electrons. The angular channels are labeled by (ℓ, m) quantum numbers that determine the multichannel expansion. **[MEASURED]** Paper 15 [9] recovers 96.0% of the H_2 dissociation energy using this geometry.

Level 5: LiH via composed geometries. For core–valence systems, the total geometry is a fiber bundle $G_{\text{total}} = G_{\text{nuc}} \rtimes G_{\text{core}}(R) \rtimes G_{\text{val}}(R)$, where each electron group occupies its own natural geometry. The angular labeling (ℓ, m) appears independently in each fiber. Paper 17 [10] applies this to LiH and BeH^+.

[OBSERVATION] The invariance of the angular sector across all five levels is not imposed by hand: it tracks the symmetry each geometry retains (m exactly wherever axial symmetry survives; ℓ exactly wherever the full SO(3) does—the two-center problem breaks SO(3) to axial SO(2), replacing ℓ by a separation index, as §V records). The packing construction provides the combinatorial substrate for this angular factor.

The node structure as combinatorial invariant

The preceding survey reveals a structural feature that merits explicit **[OBSERVATION]** comment. The same discrete quantum number skeleton $\{(\ell, m) : 0 \le \ell,\ -\ell \le m \le \ell\}$—with its degeneracy $2\ell+1$ and selection rules—admits multiple conformally distinct continuum embeddings. The five geometries catalogued in Table II are different Riemannian manifolds with different metrics, curvatures, and coordinate singularity structures. S^3 is compact with constant positive curvature; prolate spheroidal coordinates are non-compact with coordinate singularities at the foci; the hyperspherical manifold S^5 is compact but six-dimensional (for two electrons); the composed fiber bundle is a product of manifolds with different dimensionalities at each level. Yet all of them share a common combinatorial core: the (ℓ, m) node structure with its labeling, degeneracies, and angular momentum coupling rules.

This inverts a standard universality argument familiar from statistical mechanics and lattice field theory, where many different lattice discretizations (square, triangular, hexagonal) flow under renormalization to the same continuum field theory. The standard conclusion is that the continuum description is fundamental and the lattice is an approximation. Here the relationship runs in the opposite direction: one discrete node structure admits many continuum realizations, none of which is privileged. The lattice is the invariant; the continuum geometries are projections.

We state this as a mathematical observation, not a metaphysical claim. The conformal equivalences between the continuum limit of the graph Laplacian and each continuum operator are established case by case: **[SYMBOLIC PROOF]** Paper 7 [6] provides 18 symbolic proofs of the Fock projection chain for the S^3 embedding (the graph-to-continuum convergence itself is only indicated numerically, through $n_{\max} = 30$, and is not established), and the subsequent papers verify the angular-sector recurrences and benchmark each geometry in the hierarchy numerically. The framework is demonstrably conformally equivalent to standard continuous quantum mechanics at every level where it has been tested. Whether one regards the discrete or continuum description as more fundamental is a choice of interpretation, not a consequence of the mathematics.

An analogy to isospectral manifolds is apt. Kac's celebrated question—"Can one hear the shape of a drum?"—established that different Riemannian geometries can share spectral data (the answer, as Gordon, Webb, and Wolpert showed, is no: non-isometric manifolds can be isospectral). **[OBSERVATION]** The present result is stronger in a specific sense: the geometries in Table II share a labeled node structure—quantum number assignments and per-shell degeneracies, with the (ℓ, m) selection rules shared exactly wherever the corresponding symmetry survives (m in every row; ℓ in every row except the two-center Level 2, where SO(3) breaks to axial SO(2)). The spectra themselves differ across rows; what is shared is combinatorial structure, which is richer than a list of eigenvalues and different in kind from isospectrality.

GRAPH TOPOLOGY AND THE BRIDGE TO DYNAMICS

Vertex set

The packing defines a graph $G = (V, E)$ with vertex set

$$V = \left\{(n, \ell, m, s) : 1 \le n \le n_{\max},\ 0 \le \ell < n,\ -\ell \le m \le \ell,\ s \in \{\uparrow, \downarrow\}\right. \tag{7}$$

with cardinality $|V| = \sum_{n=1}^{n_{\max}} 2n^2 = \frac{1}{3} n_{\max}(n_{\max} + 1)(2n_{\max} + 1)$ for a lattice truncated at $n_{\max}$: each shell n contributes its $2n^2$ spin-doubled capacity, summed over $1 \le n \le n_{\max}$. (The top shell alone holds $2n_{\max}^2$; the cumulative count is the sum.) **[SYMBOLIC PROOF]** This spin-doubled vertex count is verified for all $n_{\max}$ symbolically (`tests/test_paper0_vertex_count.py`) and by enumeration (`tests/test_dirac_lattice.py`, which works in the (n, κ, m_j) labelling; the spinless production lattice holds $\sum n^2$ nodes).

Edge set

Edges encode geometrically adjacent transitions:

- **Radial edges** ($\hat{T}_\pm$): connect (n, ℓ, m, s) to $(n \pm 1, \ell, m, s)$ at fixed angular quantum numbers (ℓ, m). These correspond to radial excitation and de-excitation.
- **Angular edges** ($\hat{L}_\pm$): connect (n, ℓ, m, s) to $(n, \ell, m \pm 1, s)$ at fixed n and ℓ. These correspond to magnetic sublevel transitions within a shell.

The full graph topology may also include edges connecting states with different ℓ (e.g., ℓ-changing transitions driven by non-spherical perturbations). However, for the spherically symmetric Coulomb problem, the Hamiltonian is block-diagonal in ℓ: the spherical symmetry decouples the ℓ sectors, so that the graph Laplacian decomposes into independent ℓ-blocks (m is mixed within a block by the azimuthal edges; the production graph has no ℓ-changing edges — corrected 2026-09-03 from "(ℓ, m) blocks"). This block-diagonality is a consequence of the potential's symmetry, not a topological constraint on the graph itself.

Spectral convergence

The graph Laplacian $L = D - A$ defines a discrete Hamiltonian

$$H = K_{\mathrm{vac}}\, L\,, \tag{8}$$

where $K_{\mathrm{vac}} = -1/16$ is the kinetic scale factor that maps dimensionless graph eigenvalues to physical energies in atomic units. **[OBSERVATION]** This constant's value $1/16$ *coincides* with a conformal-geometry quantity of the Fock projection (Paper 7 [6]): the squared coupling between adjacent n-shells in the Gegenbauer eigenbasis of the S^3 Laplacian is $c^2(n,l) = \frac{1}{4}[1 - l(l+1)/(n(n+1))]$ (Paper 7, corrected 2026-09-03 from a printed 1/16 prefactor), and the quantity that coincides with 1/16 is the inverse Fock Jacobian $1/\Omega^4(0)$ with $\Omega(0) = 2$, equivalently $c^2(n,0)/4$. This numerical coincidence between the energy-matching scale and the geometric 1/16 is an observation, not a derivation (Paper 7): the matching scale and the conformal quantity are computed from disjoint inputs, with no known mechanism connecting them.

[MEASURED] The edges change n (radial, $T_\pm$) or m (azimuthal, $L_\pm$) but never ℓ, so the Laplacian splits into ℓ-blocks; the $\ell = 0$ block is a path graph in n whose top eigenvalue tends to 4, while the full graph's largest Laplacian eigenvalue converges to $\lambda_{\max} \to 8 = 2d_{\max}$, twice the coordination number, attained in a mid-ℓ block ($\ell = 11$ at $n_{\max} = 30$) (`tests/test_trunk_qa_kappa.py`, `tests/test_paper7_graph_convergence.py`; 2026-09-03 correction of an earlier "within each (ℓ, m) sector" statement). **[INTERNAL THEOREM]** The saturation has a closed-form rate. Because no edge changes ℓ, the Laplacian is block diagonal in ℓ and each block is the grid graph $P_{n_{\max}-\ell} \times P_{2\ell+1}$ (radial × azimuthal paths), so the whole spectrum is

$$\mathrm{spec}(L) = \left\{\, 2-2\cos\frac{j\pi}{n_{\max}-\ell} \;+\; 2-2\cos\frac{k\pi}{2\ell+1} \,\right\}, \tag{9}$$

Each block's top eigenvalue is $4+2\cos\frac{\pi}{n_{\max}-\ell}+2\cos\frac{\pi}{2\ell+1}$, so the deficit is $\min_\ell\big[4\sin^2\frac{\pi}{2(n_{\max}-\ell)} + 4\sin^2\frac{\pi}{2(2\ell+1)}\big]$; minimising its leading form $\pi^2[(n_{\max}-\ell)^{-2} + (2\ell+1)^{-2}]$ under the block constraint gives the constant in closed form,

$$2d_{\max}-\lambda_{\max} \;=\; \frac{C}{n_{\max}^2}+o\big(n_{\max}^{-2}\big), \qquad C \;=\; \frac{\pi^2}{4}\big(2+2^{1/3}\big)^2\big(1+2^{-2/3}\big) \tag{10}$$

attained at $\ell/n_{\max} \to 1 - 2/(2+2^{1/3}) = 0.386488$ (measured argmax $\ell = 11$ at $n_{\max} = 30$, 27 at 70). Every finite sample lies *below* C (verified at all 591 integer cutoffs $10 \le n_{\max} \le 600$) — $C_n = 42.61$ at $n_{\max} = 320$, 42.70 at 1000, 42.73 at 4000 — so a finite sample always understates C. The approach is *not* monotone, however: there are 195 decreasing steps in that range, e.g. $C_{20} = 40.7285 > C_{21} = 40.6470$ (corrected 2026-09-04; "monotone from below" was asserted from a doubling grid that happens to be monotone). It is the one-sided bound, not monotonicity, that makes every sample an understatement, and the value printed here before 2026-09-03, 42.6, was the $n_{\max} = 320$ sample rather than the limit (`tests/test_paper1_block_spectrum.py`). Finally, with $H = K_{\mathrm{vac}}L$ and K_{vac} fixed by the match $K_{\mathrm{vac}} = E_{\mathrm{target}}/\lambda_{\max} = -0.5/8$, the ground-state eigenvalue of H is then -0.5 Ha by construction. The convergence of the spectral bound is the content; the reproduction of the target is not. The **[SYMBOLIC PROOF]** (ℓ, m) angular structure is exact at every finite $n_{\max}$—it is encoded in the graph topology, not approximated by mesh refinement.

The conformal equivalence between the continuum limit of this graph Laplacian and the Laplace–Beltrami operator on the unit S^3 (via Fock's stereographic projection) is developed in Paper 7 [6], **[SYMBOLIC PROOF]** where an audit of 18 symbolic checks (several limiting-case or definitional) validates the conformal geometry (the convergence itself is only indicated numerically, not established, through $n_{\max} = 30$; the rate is a separate result). The energy eigenvalues and dynamics built on this graph are developed in Paper 1 [11] and the archived Paper 6 [12].

SCOPE AND LIMITATIONS

What the construction provides

- ✓ The angular quantum number labeling (ℓ, m) and the per-shell degeneracy $2\ell + 1$, as a direct consequence of uniform-density isotropic packing.
- ✓ A graph topology (V, E) whose vertices carry quantum number labels and whose edges encode allowed transitions.
- ✓ The structural correspondences $n \leftrightarrow$ cumulative grouping and spin multiplicity $\leftrightarrow$ orientability, with their proper epistemic status identified.

What the construction does not provide

× Energy eigenvalues $E_n = -1/(2n^2)$. These require the graph Laplacian together with the matched kinetic scale $K_{\rm vac}$, whose value is an Observation rather than a derivation (Papers 1, 7).

× Wavefunctions $R_{n\ell}(r)$ or $Y_{\ell m}(\theta, \phi)$ in their functional form. The packing produces the labels, not the functions.

× Time evolution and dynamics. These require the Hamiltonian as an operator, developed in the archived Paper 6 [12].

× The fine structure constant or any coupling constants. The construction is purely kinematic.

Choices beyond the axioms

As noted in Sec. , the construction involves structural choices—concentric shells, linearly spaced radii ($r_k = kd_0$), area-proportional counting, and the fundamental area $\sigma_0 = \pi d_0^2/2$—that are natural and mutually consistent but not uniquely determined by the two axioms. **[OBSERVATION]** The robustness of the (ℓ, m) result rests on the observation that any isotropic, uniform-density, concentric packing produces the same angular capacity $2k-1$ per shell; the specific radial spacing affects only how shells are ordered, not their angular content.

DISCUSSION

Historical context

The connection between geometry and quantum mechanics has been recognized since the earliest days of quantum theory. Bohr [5] imposed quantization as a geometric constraint (angular momentum $L = n\hbar$ on circular orbits, later reinterpreted by de Broglie as an integer number of wavelengths around the orbit). Fock [1] discovered that the hydrogen momentum-space problem maps onto S^3 via stereographic projection, revealing the SO(4) dynamical symmetry that explains the "accidental" n^2 degeneracy. Bargmann [2] derived the same symmetry algebraically. Barut and Kleinert [3] extended the symmetry analysis to transition probabilities via non-compact dynamical groups. Biedenharn [4] provided a comprehensive treatment of the angular momentum algebra underlying the (ℓ, m) structure.

The present work operates at a different level: rather than analyzing the symmetries of the Schrödinger equation, it constructs the angular labeling from geometric–combinatorial reasoning that is logically independent of the operator formalism. The result is the same (ℓ, m) basis—as it must be, since both approaches reflect the same underlying rotational symmetry—but obtained by a different route.

The kinematic–dynamic distinction

The packing construction produces the *kinematic* structure of quantum mechanics: the state space, its labeling, and its combinatorial organization. It does not produce the *dynamics*: the Hamiltonian, the energy spectrum, or time evolution. This separation is a feature of the construction, not a limitation. It shows that the angular quantum numbers are determined by geometry alone, independently of the specific force law or potential.

In the GeoVac series, dynamics are recovered when the graph Laplacian is equipped with the kinetic scale $K_{\rm vac} = -1/16$ and node weights encoding the Coulomb potential. Paper 7 [6] shows that the continuum limit of this graph Laplacian is conformally equivalent to the Laplace–Beltrami operator on S^3 (the convergence itself is only *indicated* numerically through $n_{\max} = 30$, not established — corrected 2026-09-04 to match Paper 7 and §VI above, both of which already stated it correctly), with the $1/r$ Coulomb potential emerging as the coordinate distortion from stereographic projection rather than an independent force law.

Scope of the geometric analogy

The construction shows that information-theoretic reasoning under geometric constraints *can* reproduce the angular quantum number structure. It does not show that this is the *only* way to obtain these labels, nor does it establish that quantum mechanics *must* have a geometric origin. The angular momentum algebra $[\hat{L}_i, \hat{L}_j] = i\epsilon_{ijk}\hat{L}_k$ is a consequence of rotational symmetry regardless of whether one arrives at it through operator theory or geometric packing. What the construction demonstrates is that the *state-counting* aspect of angular momentum—the degeneracy $2\ell+1$—has a purely geometric derivation.

CONCLUSIONS

The angular momentum quantum numbers (ℓ, m) emerge from uniform-density packing on isotropic shells: shell k carries $2k-1$ angular slots, isomorphic to the spherical harmonic basis $Y_{\ell m}$ with $\ell = k-1$. This result follows from two axioms (binary distinguishability and geometric indifference) together with natural structural choices (concentric shells, uniform density). It is independent of the Schrödinger equation, operator postulates, or any specific physical system.

The principal quantum number n corresponds to cumulative grouping of the first n shells, yielding n^2 orbital states. The factor of 2 corresponds to the orientability of the compactified packing manifold. These are structural correspondences—genuine mathematical observations that match hydrogen's quantum numbers—but they depend on identifications beyond what the two axioms alone determine.

The (ℓ, m) angular sector constructed here recurs as the universal angular cross-section inside every natural geometry developed in the GeoVac series, from S^3 for single-electron atoms [6] through composed fiber bundles for core–valence molecules [10]. When equipped with a graph Laplacian and kinetic scale, this packing becomes the foundation for a computational framework that has been applied to H, He, H_2^+, H_2, HeH^+, **[MEASURED]** LiH, and BeH^+, with a saturation deficit of 0.57% for the hydrogen ground-state bound on the production graph at $n_{\max} = 30$ (a by-construction quantity, §VI; 0.11% at $n_{\max} = 70$, `tests/test_paper7_graph_convergence.py`; the continuum chain itself is exact) and 5.3% for the LiH equilibrium geometry (BeH^+ is markedly worse: R_{eq} 3.25 vs ~ 2.48 bohr with the ab-initio pseudopotential; HeH^+ is treated at a fixed bond length and recovers 93% of D_e adiabatically but only $\sim 5\%$ in the consistent 2D-variational solver; Paper 17), benchmarked against exact and experimental values.

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